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TRACK 2: MEDICAL APPLICATIONS OF MULTIMEDIA

Development of framework by combining CNN with KNN to detect Alzheimer's disease using MRI images

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Abstract
Alzheimer's disease (AD) is an irremediable, irrecoverable brain illness without proper treatment. Therefore, AD recognition is essential for precluding and dominating its advancement. The conventional method considers the patient history, MRI, and empirical tests to detect AD. This method is effective, but it lacks accuracy. The primary objective is to create an efficient framework for timely AD identification from MRI images. This work combines a Convolutional Neural Network (CNN) with the K-nearest neighbor (KNN) to detect AD from 6400 MRI images. The dataset consists of four types of AD MRI images such as Moderate Demented (MOD), Mild Demented (MID), Very Mild Demented (VMD), and Non-Demented (ND). The dataset was split into training (80%) and validation (20%) sets. The CNN was utilized to extract the informative features automatically from the MRI dataset images, and these extracted features are used to train and validate the KNN model. The CNN-KNN framework is accessed utilising the Receiver Operating Characteristic (ROC) curve, stratified K-fold, Cohen's Matthews Correlation Coefficient (MCC) and Kappa Coefficient (CKC). This method achieved average 99.58% accuracy, 99.63% precision, 99.31% recall, 99.43% F1-score, 99.31% MCC, and 0.9931 CKC. Furthermore, three deep CNNs (DCNNs), ResNet50, VGG16, and MobileNetV2, were employed to detect AD and compare their performance with CNN-KNN. The proposed CNN-KNN method's performance has significantly improved compared to the DCNN and literature.

Keywords CNN-KNN · VGG16 · MobileNetV2 · ResNet50 · Stratified K-fold

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1 Introduction

AD is a neurological brain illness that impairs people's mental health. In severe conditions, patients lose their capacity to think, read, and write and may not even remember family members. AD is dementia that appears in people who show symptoms of beta-amyloid in the brain and may require a full-time caretaker [43]. Millions of people worldwide continue to suffer from AD, despite efforts to discover an effective treatment for it [58]. Early AD detection in patients is essential for the care and clinical trials. Early detection and appropriate treatment can prevent the patient from progressing to severe Alzheimer's. The contraction of the brain cortex is one of the most common factors, and normal contraction occurs primarily in the hippocampus. This area is responsible for thinking and remembering daily tasks. A diminish in the hippocampal region controls to shrinkage of the cerebral cortex and swelling of the ventricles [43].

As per the National Institutes of Health (NIH) and the Alzheimer's Association [3], 50–80% of dementia patients progress to AD [40]. The majority of patients are diagnosed after the age of 65 to 70. However, early recognition and prognosis may help reduce the influence of illness development. The major obstacle for Alzheimer's specialists is that no trustworthy medication for AD is available [52]. Despite this, contemporary AD medications can reduce or halt the course of symptoms. As a result, detecting AD in its early stage is crucial [63]. A computer-aided system (CAD) is employed for precise and initial AD recognition to evade expensive care that is forecast to be enhanced rapidly [11]. Current approaches rely on manual feature extraction, a time-consuming, subjective, and technical skill of experts. As a result, deep learning (DL), particularly CNNs, is an excellent method for addressing these issues [44]. CNN can improve efficiency and has proven significant effectiveness in AD detection and does not require manual feature extraction because it extracts the features automatically [64].

Timely screening of AD for medical assessment is arduous by high relativism, specific variance in cognitive estimates, and the protein biomarkers assessment mainly utilised for AD staging. Even though there is still no remedy for AD, early diagnosis of dementia via MRI analysis may aid in the development of precautionary measures to slow disease progression. Due to this, an accurate AD prediction system is vital to help clinicians. The motivation and objective of this work are to create a software solution that will assist doctors in identifying AD. Due to this, an efficient CNN-KNN-based system was developed to assist physicians or doctors in detecting AD. This system can also help learn new concepts, AD diagnosis, disease behaviour prediction, disease estimation, and medication use. This current proposal may help physicians detect illness more quickly.

In this paper, the authors developed an automated ML framework for recognizing four stages of AD by combining CNN and KNN. The study attempts to generate a new way by combining the CNN method with the ML classifier (KNN). In the CNN-KNN, CNN is used for extracting informative features from the images and KNN classification of these features. Because a quick disorder predicting system needs, this research topic must be addressed. On the MRI datasets, the authors used a simple random sample approach. CNN has demonstrated exemplary performance and generalization abilities, which may play a role in its use in various applications. Despite its outstanding performance, CNN poses unique challenges, such as balancing effectiveness and processing times. To ensure predictive performance, the authors investigate various network structure hyperparameter combinations. The authors explore different variants in input image size, convolutional layers with multiple filter sizes, optimizers, effective use of batch normalization layers, and so on, with their own hyperparameters.

When using an MRI image in CNN, filters or kernels generate feature maps. The feature maps highlight various visual elements in the original MRI image. As a result, the size of the filter selection may enable cost-effectiveness for the proposed system. This research will concentrate on using suitable learning rate regard and vector reduction of input data vectors. With Max pooling levels, the CNN kernel size remains small for optimal feature extraction. The CNN extracted features were then used to classify AD using the KNN classifier. The authors produced excellent outcomes compared to previously reported work by evaluating the MRI images to identify the AD stages.

Although other articles employed CNN in conjunction with SVM or RF for image classification, none of the authors used CNN in conjunction with KNN for AD detection using MRI images. A unique method combining a CNN with a KNN classifier is proposed to detect AD using four types of MRI images: mild dementia (MID), moderate dementia (MOD), non-depression (ND), and very mild dementia (VMD). The CNN has extracted the informative features from MRI images. These extracted features were employed to train the KNN model, and performance was evaluated using confusion matrix, ROC curve, MCC, CKS, and stratified K-fold cross-validation techniques. The proposed CNN-KNN approach increases classification accuracy and tackles the class imbalance problem inherent in multiclass classification situations. The proposed CNN-KNN algorithm can categorize and detect four phases of AD: MID, MOD, ND, and VMD. The contribution of this work is as follows:

- The paper proposes a CNN that extracts informative features from MRI scans to identify AD patients. The KNN classifier is utilized to classify the CNN extracted features. In other words, a CNN-KNN approach is presented.
- Three states of art models, namely ResNet50, VGG16, and MobileNetV2, were developed to compare the performance of the CN-KNN model.
- The CNN-KNN, ResNet50, VGG16, and MobileNetV2 models were trained and tested on a dataset of 6400 MRI scans, whereas recent studies on creating a similar type of system have used relatively small datasets.
- The achievement was estimated through accuracy, ROC curve, precision, recall, F1-score, MCC, CKS, and stratified K-fold cross-validation approaches of the CNN-KNN model.
- The CNN-KNN and DCNN were built using the Python.

The following is how the paper is organized: Section 2 discusses the related works. Section 3 discusses material and methods. Section 4 presents the experiment and analyses the outcomes. Section 5 discusses the comparative analysis of the present work with literature reported work. Section 6 concludes the paper and discusses future work.

2 Related work

Recent research suggests considerable improvement in identifying and classifying AD using multimodality data. Positron emission tomography (PET), X-rays, MRI, computed tomography (CT), and the patient's clinical records are all included in multimodality data [51]. Although MRI is more effective than CT scans in identifying AD when utilizing traditional machine learning (ML), there are numerous significant difficulties associated with these techniques [43]. Significant progress has been made in the automatic classification of AD using various methods in recent years. Various ML approaches, such as KNN [8, 22, 23, 31, 41, 54], Random Forest (RF)

[8, 9, 18, 41, 45, 54, 62]), SVM [6, 13, 17, 23, 36, 41, 54], Decision Tree (DT) [19, 41], Deep Neural Network (DNN) [12], CNN [10, 20, 28, 37, 48, 53, 57, 65], DCNN [7, 21, 25, 26], and dynamic connectivity networks (DCN) [27], have been employed to detect AD using MRI images. Moreover, various studies combined CNN with other ML models, such as the CNN-long short-term memory (LSTM) [24], CNN-RNN [1], CNN-RF [56], CNN-SVM [56], CNN-DT [56], and CNN-AdaBoost [56].

Nozadi et al. [45] presented a pipeline for performing group classification utilizing learned features using semantically labelled positron emission tomography (PET) pictures. An annotated MNI template was combined with every patient-specific PET scan using a deformable multi-modal PET-MRI registration process, and areas of interest were retrieved for AD diagnosis. SVM and RF classifiers were employed for classification. The RF classifier achieved the highest 91.2% accuracy for AD versus NC when tested on 660 participants from the ADNI database. They have classified only two AD and NC, and performance is not as per our work. Gunawardena et al. [17] employed SVMs and CNN and obtained 84.4% accuracy, 95.3% sensitivity, and 71.4% specificity for SVM, whereas the CNN model achieved 96% high accuracy and 96% sensitivity, and 98% specificity. The accuracy is less compared to our work.

Liu et al. [36] introduce approach for identifying AD that utilises spectrogram information derived from speech data. This method can be helpful to patients to know the illness development of patients at an early phase, allowing them to take preventive measures. The ML approaches such as AlexNet and GoogLeNet applied and obtained average classification accuracies of 91.40% and 93.02%, respectively. The performance is poor compared to our work. Arco et al. [6] proposed an SVM-based method integrated with Searchlight and Principal Component Analysis (PCA) to detect AD using the ADNI dataset. The Searchlight method performed well compared to the PCA and achieved a maximum accuracy of 80.9%. Helaly et al. [20] classified four stages of AD using deep learning (DL) approaches (CNN and VGG16). Moreover, distinct binary medical image categories were used for each two-pair class of AD stages. The CNN and VGG19 methods achieved classification accuracies of 93.61% and 95.17% for 2D and 3D multiclass AD for CNN, respectively, whereas VGG19 achieved a classification accuracy of 97% for multiclass AD. The classification accuracy is less than our work. Sarraf et al. [57] proposed a CNN (LeNet-5) model to recognize Alzheimer's brain from the normal healthy brain and categorised MRI images of Alzheimer's subjects from control subjects, with 98.84% accuracy. This investigation indicates that using CNN to retrieve shift and scale invariance followed by DL classification is the most efficient technique for distinguishing clinical data from healthy data in fMRI. This strategy also allows broadening methodology to foresee more complex systems.

Wang et al. [65] proposed a CNN with the rectified linear unit (ReLU), sigmoid, leaky ReLU, average, max, and stochastic pooling to test the model and achieve 97.65% accuracy. Jain et al. [26] used the VGG16 model to detect AD and achieved an accuracy of 95.73%. Rallabandi et al. [55] analysed 1167 MRI images from the ADNI database. They used the FreeSurfer method to assess regional cortical thickness in the left and right hemispheres (68 features) and then used these 68 features to build models. They discovered that a non-linear SVM had the highest 0.75 sensitivity, 0.77 specificities, 0.72 F1-score, 0.69 Kappa-statistic, 0.71 MCC, 0.76 ROC-AUC, and overall 75% accuracy in categorising all four categories utilising 10-fold cross-validation. This method's performance is poor compared to our approach. Al-Shoukry et al. [2], Gao et al. [14], and Arafa et al. [5] briefly reviewed some of the relevant works of literature on AD and explored how DL can assist scientists in identifying the illness stages. Pan et al. [47] suggested the CNN-EL approach using MRI to select patients with MCI or AD. A significant number of CNN were trained for each binary classifier using a set of coronals, sagittal, or transverse MRI slices; these CNN models were

combined into a single group. The outcome was assessed using the stratified 5-fold cross-validation method. The highest accuracy rate for classifying AD was 0.84, less than our presented work. Muhammed Raees et al. [42] proposed SVM and DNN methods by using a database of 111 subjects in Mild Cognitive Impairment (MCI), AD, and Normal classes to identify AD stages and high accuracy of about 80–90% on AD prediction. This method has poor performance in identifying AD stages. Odusami et al. [46] suggest a DL-based framework for determining early MCI, MCI, late MCI, and AD. The ADNI fMRI dataset with 138 subjects was utilised for the assessment. On the early MCI vs AD, MCI vs AD, and MCI vs early MCI scenarios, the finetuned Resnet-18 obtained a 99.99%, 99.95%, and 99.95% accuracy, respectively. This system performs well but only classifies two classes individually, not all four categories.

According to the above publications, numerous studies on discovering AD have been initiated. The image preprocessing method is used to determine malady regions and categorise them using various DL models. It has also been found that there is room for an investigation into developing a method to detect AD in its early stages with high precision. The methodology must successfully see early-stage symptoms of the disease. The investigation seeks to obtain new information about this dilemma by carefully examining it with systematic concepts. The authors tried the technological applied research method in this paper. The primary aim of this study is to upgrade existing methodologies by customising CNN structures to retrieve informative features and ML classifiers to accurately classify AD. The authors used a simple random sampling method with the Kaggle MRI dataset. Because CNN-KNN frameworks can assist find new solutions, which will help to solve the above problem. When handling vast amounts of information, DL can outclass traditional ML methods. Manual feature extraction is used in ML, but it is hard to retrieve features manually from large datasets. However, DL balances well with vast amounts of information. Furthermore, DL, particularly CNN, relies less on feature engineering and automatically extracts features from the images.

3 Material and methods

The entire system structure for detecting AD using MRI images is depicted in Fig. 1. The pictures of the MRI dataset were divided into four types. The raw MRI images were first processed through the image preprocessing technique, which included image rescaling. The

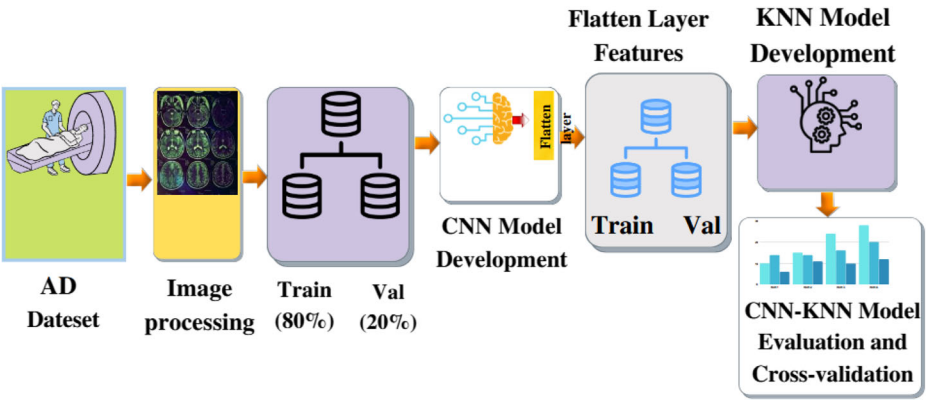


Fig. 1 The overall system framework of the proposed AD identification system

preprocessed data set was then divided into a training set and a validation set, and the CNN was trained with the training data. The CNN extracted features or flatten layer output is further subdivided into training and validation features. These features were used to train and validate the KNN model. The suggested system's effectiveness was evaluated using the following methods/metrics: confusion matrix, accuracy, ROC curve, precision, recall, F1-score, CKS, and MCC. Furthermore, the CNN-KNN model was cross-validated utilising 10-fold, 15-fold, and 20-fold cross-validation technique.

3.1 Alzheimer's disease (AD) MRI dataset

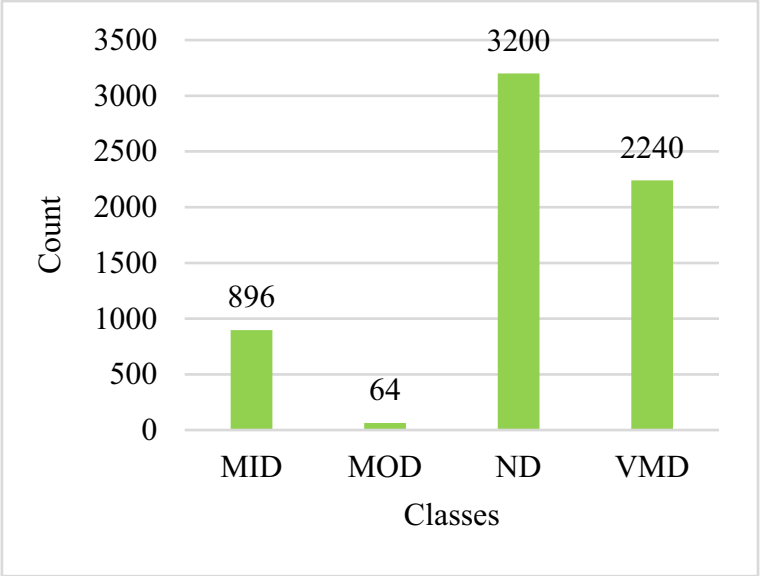
The MRI images dataset available publically on Kaggle is used for the current study [4]. The main reason for posting this Dataset publically is to develop a highly accurate model that predicts the AD stages. It consists of four types of AD MRI images such as Mild Demented (MID), Moderate Demented (MOD), Non-Demented (ND), and Very Mild Demented (VMD). The Dataset contains a total of 6400 MRI images, as shown in Fig. 2a, and four AD disease images, as shown in Fig. 2b. The MID has 896 images, MOD has 64, ND has 3200, and VMD has 2240. The data is collected from several websites/hospitals/public repositories. This is an imbalanced dataset. AD has varying degrees of severity. In the VMD phase, the patient begins to forget where they placed their belongings, other people's names, and so on. It is difficult to recognise using cognitive ability tests. In the MID phase, patients don't recall what they're saying, can't find their way to their desired location, and have lost focus and work abilities. At this point, patients may even be unaware that they have lost memory. It can be discovered at this stage through cognitive testing. In the MD phase, the patient begins to forget recent efforts and essential old histories, has difficulty calculating the expenditure, finds it difficult to go out on their own, and loses empathy.

3.2 Image pre-processing

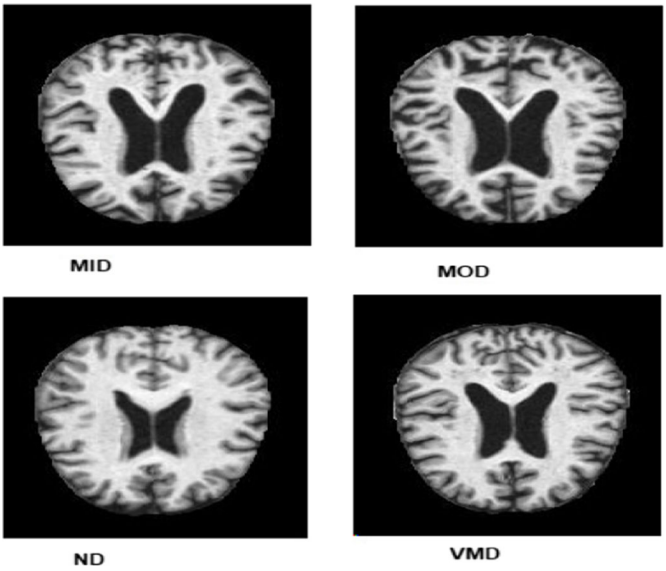
Image pre-processing intends to optimize image data by removing undesired discrepancies or boosting certain image features that will benefit subsequent processing and analysis tasks. Sometimes images may have different sizes, so an image scaling method is needed to scale the images. In this work, the MRI images were resized to 224 × 224 pixels in size. As a result, the dataset was ready to be fed into the CNN-KNN framework and used to train the framework.

3.3 Feature extraction using convolutional neural network (CNN)

Most methods used CNN, DCNN, and ML classifiers to identify AD. The feature extraction process is critical. Some of the researcher's features were extracted manually and then given to the ML classifiers to classify. Here, the authors used CNN as a feature extractor to extract informative features from MRI scans. CNN is helpful when dealing with a vast amount of pictorial data. The CNN's basic structure comprises a convolutional layer (CL), batch normalization (BN), pooling layer, and dense layer, as shown in Fig. 3 [32, 33]. The proposed model employs four convolutional blocks, namely, CB-1, CB-2, CB-3, and CB-4. Each convolutional block contains CL, BN, and Max Pooling layers. The CL helps extract various features via the kernel filtering process. The network parameters were reduced via convolutional weight sharing and pooling operations. The pooling layer further isolates the original parts, drastically reducing the training parameters. In CNN, the input MRI image is perpetuated, blocking at a predetermined layer, and the results of that layer have been used as features for the next layer. CNN has effective, racist and discriminatory feature



(a)



(b)

Fig. 2 **a** Total MRI images in the dataset including all classes and **b** dataset images of four classes

extraction characteristics and can estimate classes on which CNN has never been trained. Figure 3 depicts the CNN feature extraction process. CNN allows an image to pass from the first CL to the Max Pooling layer via the BN layer, and then the Max Pooling layer output is applied to the BN layer. The same procedure is followed up with a flattened layer. There have been N number MRI

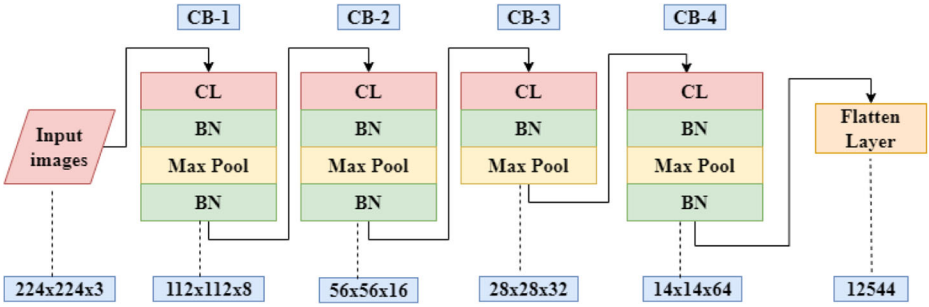


Fig. 3 Feature extraction process using CNN

images Mn, n ∈ [1, N] after the data pre-processing for feature extraction; these images were scaled to 224 × 224 and normalized to the interval [0, 1]. Furthermore, the CL with a 3 × 3 kernel size is applied for feature extraction. A rectified linear unit (ReLU), BN, and Max pooling size 2 × 2 were implemented for each convolution process. Max pooling helps to slash the preceding feature map by half.

The input image dimensions were 224x224x3. The volume shape of the first CL was 224x224x8, and the volume shape of the Max pooling layer was 112x112x8. The volume shape 14x14x64 of the fourth Max Pooling layer was flattened to feature vector 12,544 (Flatten Layer output). This feature extraction will be repeated for all photos in an N-image dataset, yielding N × 12,544 feature vectors. These extracted features were then used to predict/classify using the KNN classifier.

3.4 K- nearest neighbor (KNN)

The KNN algorithm is the supervised ML algorithm employed for classification and regression [34]. A labelled database is provided for training, and an unlabelled data point is identified based on the category of K data points closest to the unlabelled points of the neighbour. In KNN, decide the K of neighbours and compute the Euclidean distance (ED) between those neighbours. Carry the K nearest neighbours using the ED and calculate the data points within every class amid K neighbours. Allocate the recent data points to the class with the maximum number of neighbours [29]. In KNN, K is the algorithm's important parameter. The features from the flattened layer are given to the KNN model. The inputs to the KNN are the features F1, F2... Fn (n = 12,544). These features are split into training and validation datasets. The distance between validation data point F and training data point Fi, i = 1..., n, is determined using this approach [31].

$$\begin{aligned} &dE(F, Fi) = \sum_{i=1}^{n} |F - Fi| \\ &F : dE\,(F, Fi) < dE\,(F, Fi) \end{aligned} \tag{1}$$

3.5 Deep convolutional neural networks (DCNN)

In this work, three DCNN models were employed to classify AD as follows:

Residual Neural Network (ResNet): It is a DCNN that can have up to 152 layers. To skip some CLs, ResNet utilizes "gated units." ResNet also utilizes a novel design to run many more convolutional layers without increasing difficulty [32, 33, 61].

VGG-16: VGG-16 is a popular DL and 16-layer DCNN image categorization algorithm. Because of its advantages, VGG16 is often utilized in learning applications. The pretrained network from ImageNet can recognise photos into 1000 different object categories. As a result, the framework has learned detailed feature representations for a diverse set of images. The network's picture input size is 224 × 224 [32, 60].
MobileNetV2: MobileNetV2 is an advanced state-of-the-art framework for mobile visual identification, including object detection, classification, and semantic segmentation. MobileNetV2 expands on the concepts of MobileNetV1, employing depth-wise separable convolution as intelligent building blocks. However, MobileNetV2 adds two new architectural features, such as linear bottlenecks between layers and shortcut connections between bottlenecks [39].

3.6 Proposed CNN-KNN framework

Figure 4 depicts the proposed CNN-KNN framework for detecting AD. The quality of the dataset images must be upgraded using the pre-processing method. CB-1, CB-2, CB-3, and CB-4 are the four blocks of the suggested approach that makes up the sequential CNN model. Each CNN block is made up of three layers. The CL of CB-1 is provided with pre-processed photographs. The CB-1 block takes rudimentary features from MRI pictures and sends them to the CB-2 block. The CB-2 block computes and creates additional features based on the characteristics retrieved by the CB-1 block. Similarly, the CB-3 block collects features from the CB-2 block's output. This procedure is repeated until the CB-4 block. A novel technique is proposed by merging the CNN model with the KNN classifier. The CNN-KNN technique investigates how to employ the CNN model as a feature extractor and the KNN model as a classifier. The features retrieved from the CB-4 block are fed into the KNN classifier through Flatten layer, and the results reveal that the accuracy is significantly improved.

4 Results

This section discusses the performance of the DCNN and CNN-KNN models. The open-source MRI data, available on Kaggle, are used for this work.

4.1 Performance analysis

The CNN-KNN model performance was evaluated using MCC [16], CKS [16], and the stratified K-fold cross-validation techniques [32].

$$A_{AD} = \frac{TP_{AD} + TN_{AD}}{TP_{AD} + FP_{AD} + TN_{AD} + FN_{AD}} \tag{2}$$

where A_{AD} -Accuracy, TN_{AD}-Truly Negative, TP_{AD}-Truly Positive, FP_{AD}-False Positive, and FN_{AD}-False Negative

$$Precision\ (P_{AD}) = \frac{TP_{AD}}{TP_{AD} + FP_{AD}} \tag{3}$$

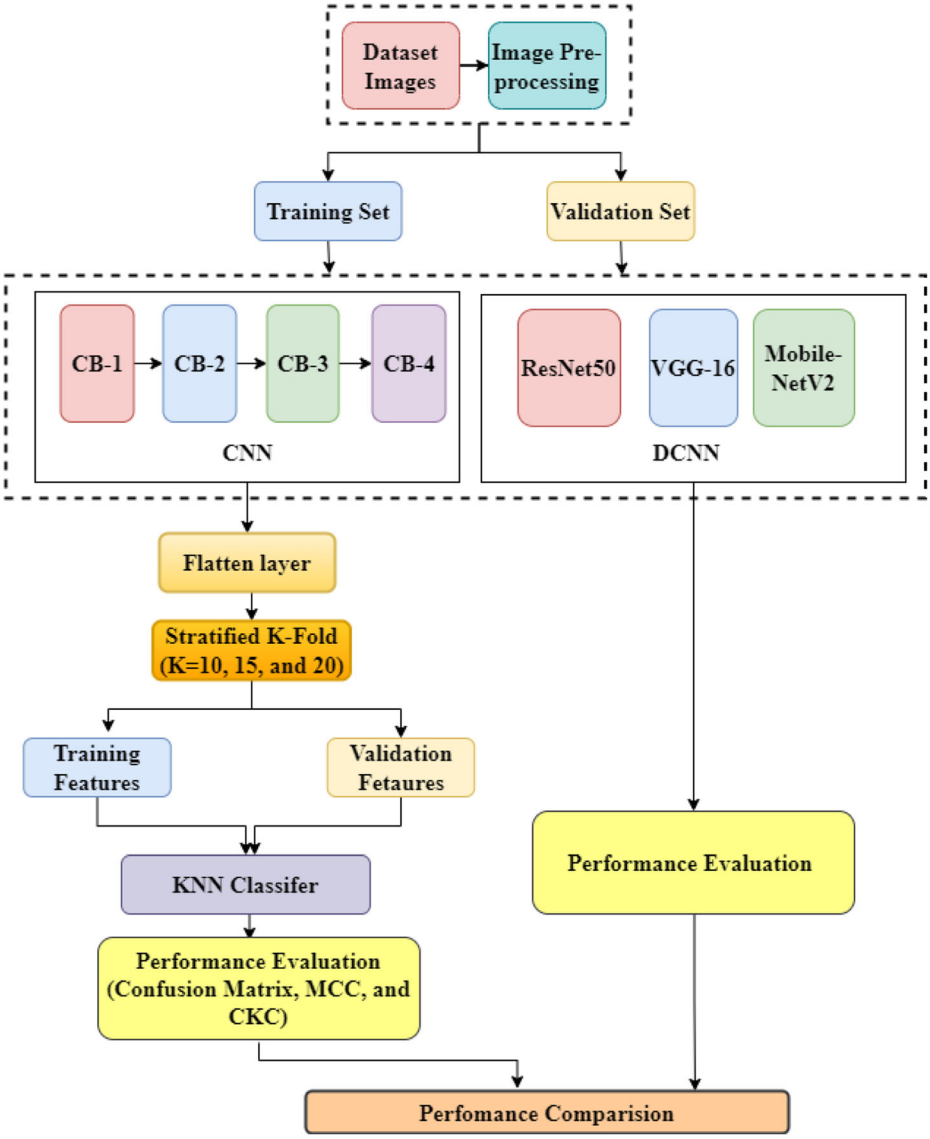


Fig. 4 Proposed CNN-KNN framework

$$Recall\ (R_{AD}) = \frac{TP_{AD}}{TP_{AD} + FN_{AD}} \tag{4}$$

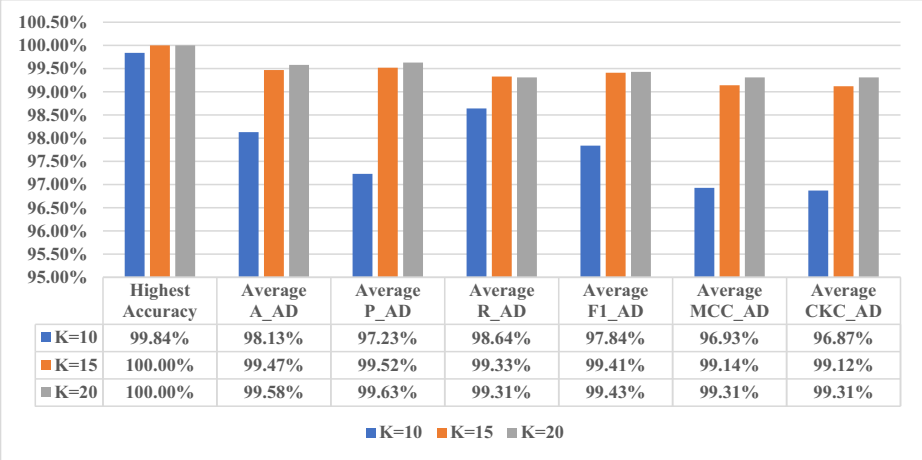
$$F1\text{–}score\ (F1_{AD}) = \frac{2}{\left(\frac{1}{R_{AD}}\right) + \left(\frac{1}{P_{AD}}\right)} \tag{5}$$

4.2 CNN and CNN-KNN performance

The Adam optimizer (lr = 0.0003) and categorical crossentropy loss function were employed in the CNN model with batches size of 32. The CNN model achieved 100% training accuracy with 0.0118 loss and 68.57% validation accuracy with 1.3 loss in 20 epochs. In the CNN-KNN model, the features extracted from the CNN model were used to train and validate the KNN model. The KNN classifier with n neighbors = 3 was utilized in this study. The performance of the CNN-KNN models was evaluated using the stratified K-fold cross-validation method as depicted in Fig. 5.

The CNN-KNN model was tested on K values equal to 10, 15, and 20. The CNN-KNN model achieved the highest accuracy of 99.84% for K, equal to 10. The model achieved an average accuracy of 98.13%, average precision of 97.23%, average recall of 98.64% and average F1-score of 97.84%. The MCC is a far more dependable statistical rate that yields a high score only if the forecast performs well for all four confusion matrix classes, according to the number of positive and negative items in the dataset [16]. The CNN-KNN model achieved an average MCC of 96.93%, demonstrating good performance. On the other hand, the Cohen kappa score (CKS) is a trustworthy statistic that may be used to assess interrater and intrarater reliability [38]. Cohen recommended interpreting the Kappa score as 0 for no agreement, 0.01–0.20 for none to a bit of understanding, 0.21–0.40 for fair agreement, 0.41–0.60 for moderate agreement, 0.61–0.80 for considerable agreement, and 0.81–1.00 for almost complete agreement [16]. The CNN-KKN model achieved an average CKC of 0.9687, which indicates practically complete agreement.

On the other hand, the CNN-KNN model achieved the highest accuracy of 100% for K, equal to 15 and 20. The model achieved an average accuracy of 99.47% and 99.58%, average precision of 99.52% and 99.63%, average recall of 99.33% and 99.31% and average F-score of 99.41% and 99.43% for K, equal to 15 and 20, respectively. The CNN-KNN model achieved average MCCs of 99.14% and 99.31% for K, equivalent to 15 and 20, respectively, which indicates remarkable performance. Similarly, the CNN-KKN model achieved average CKCs



	Highest Accuracy	Average A_AD	Average P_AD	Average R_AD	Average F1_AD	Average MCC_AD	Average CKC_AD
K=10	99.84%	98.13%	97.23%	98.64%	97.84%	96.93%	96.87%
K=15	100.00%	99.47%	99.52%	99.33%	99.41%	99.14%	99.12%
K=20	100.00%	99.58%	99.63%	99.31%	99.43%	99.31%	99.31%

Fig. 5 Performance of the CNN-KNN model

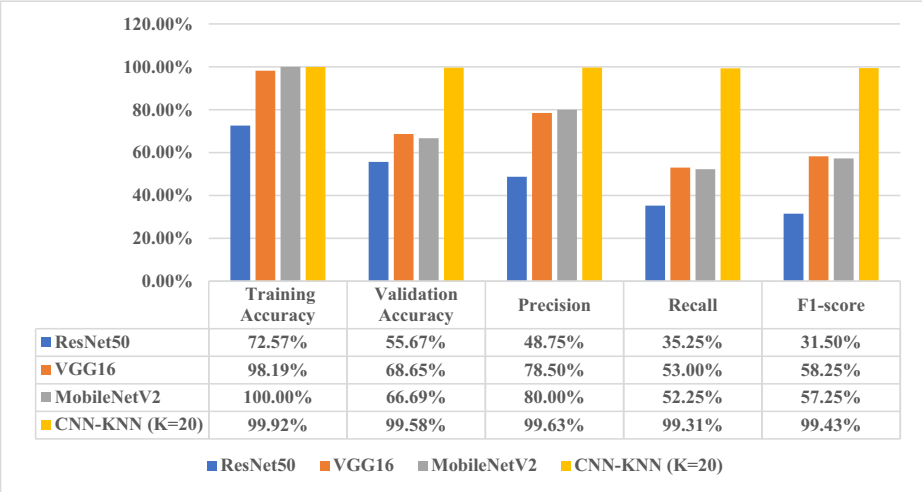
of 0.9912 and 0.9931 for K, equal to 15 and 20, respectively, which indicate almost complete agreement.

4.3 CNN-KNN and DCNN performance

Figure 6 shows the comparison of CNN-KNN with DCNN models. From Fig. 6, the training accuracies of the ResNet52, VGG16, MobileNetV2, and CNN-KNN models are 72.57%, 98.19%, 100%, and 99.92%, respectively, whereas the validation accuracies of the ResNet52, VGG16, MobileNetV2, and CNN-KNN models are 55.67%, 68.65%, 66.69%, and 99.58%, respectively. The highest training accuracy was achieved by MobileNetV2 and CNN-KNN, while the highest validation accuracy was acquired by the CNN-KNN model. Moreover, the highest precision of 99.63%, recall of 99.31%, and F1-score of 99.43% were achieved by the proposed CNN-KNN model. The performance of CNN-KNN outperformed the rest of the DCNN models for AD classification.

4.4 Receiver operator characteristic (ROC) curve of CNN-KNN

The ROC curves for CNN-KNN are shown in Fig. 7a and b. Figure 7a shows the ROC curve for the CNN-KNN model's highest 100% accuracy performance, whereas Fig. 7b shows the lowest accuracy performance of the model when the 20-fold method was applied. The ROC curve is a graphical representation of binary-class or multi-class testing skills of the ML models. The ROC curve should be closer to the upper left for improved classification performance. From Fig. 7a, it can be seen that the all-class curve on the left corner achieved an AUC of 1, which shows excellent performance. On the other hand, from Fig. 7b, the MID, MD, ND, and VMD achieved AUC of 0.97, 1, 0.94, and 0.93, respectively. This shows that CNN-KNN performs very well.



	Training Accuracy	Validation Accuracy	Precision	Recall	F1-score
ResNet50	72.57%	55.67%	48.75%	35.25%	31.50%
VGG16	98.19%	68.65%	78.50%	53.00%	58.25%
MobileNetV2	100.00%	66.69%	80.00%	52.25%	57.25%
CNN-KNN (K=20)	99.92%	99.58%	99.63%	99.31%	99.43%

Fig. 6 Comparison of CNN-KNN with DCNN models

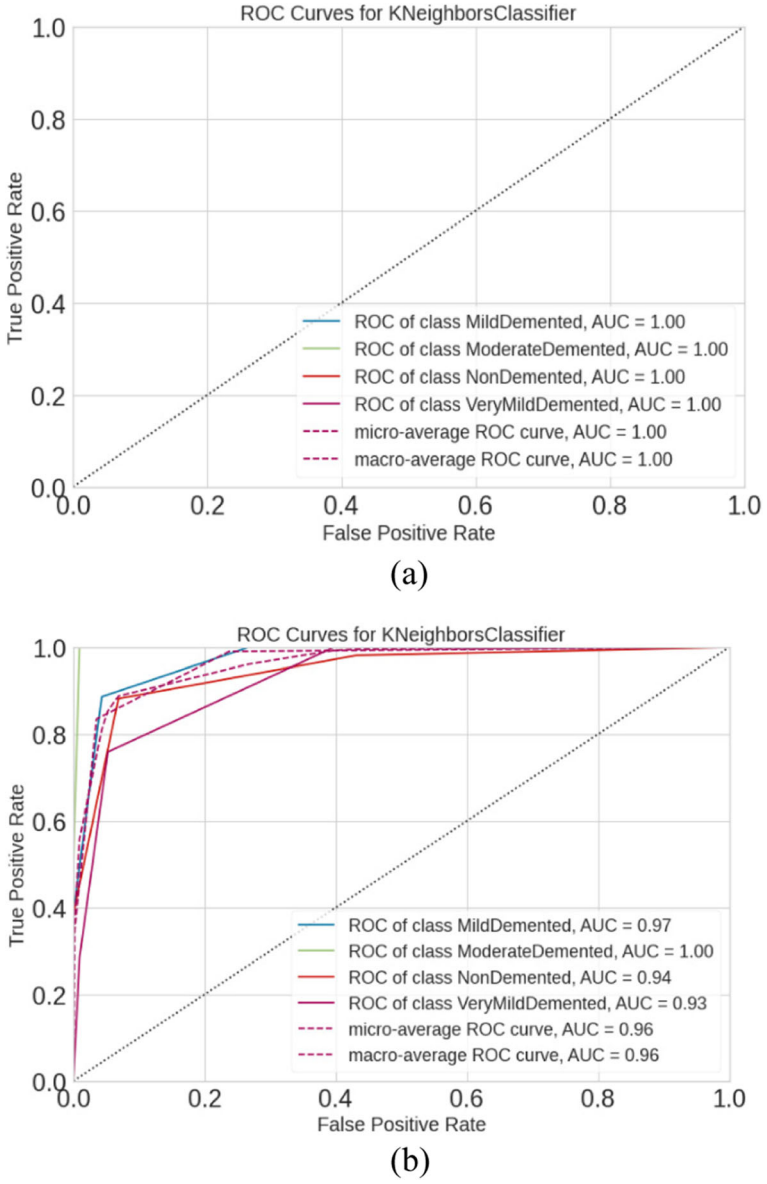


Fig. 7 ROC curve for CNN-KNN model at K = 20 (**a**) highest performance curve and (**b**) lowest performance curve

5 Discussions

This study combined the CNN with the KNN model to classify four different AD types. Several image processing with the DL approaches for identifying AD have been developed in the literature. Traditional methods can assist in determining the type of AD, but these methods are less efficient. So, there is a need to create an efficient system. Table 1 tabulates a systematic comparison of the CNN-KNN approach with other ML approaches.

Table 1 Comparisons of the CNN-KNN with other approaches

Reference	Methods	Accuracy
Kruthika et al. [31]	KNN, Naive Bayes, and SVM	96.31%
Vaithinathan et al. [62]	RF, SVM, and KNN	89.58%
Nozadi et al. [45]	SVM and RF	91.2%
Gunawardena et al. [17]	SVM and CNN	84.4% and 96%
Liu et al. [36]	AlexNet and GoogLeNet	91.40% and 93.02%
Arco et al. [6]	SVM	80.9%
Helaly et al. [20]	CNN and VGG16	95.17% and 97%
Sarraf et al. [57]	LeNet-5	98.84%
Wang et al. [65]	CNN	97.65%
Khvostikov et al. [28]	Inception-based CNN	93.3%
Jain et al. [26]	VGG16	95.73%.
Jie et al. [27]	DCN	78.8%
Rallabandi et al. [55]	SVM	75%
Pan et al. [47]	CNN-EL	84%
Muhammed Raees et al. [42]	SVM and DNN	80%–90%
Odusami et al. [46]	ResNet18	99.95%
Spasov et al. [59]	DL	87.5%
Lin et al. [35]	CNN	79.9%
Poloni et al. [50]	CNN	82.59%
Goenka et al. [15]	CNN	98.3%
Kong et al. [30]	CNN	93.5%
Proposed Work	CNN-KNN	Highest Accuracy = 100% & 99.63% Average Accuracy for K = 20

Lin et al. [35] proposed a framework to detect MCI-to-AD conversion using ADMI dataset images. First, they pre-processed the photos and then extracted local patches. These retrieved patches were employed to train a CNN to detect MCI features. FreeSurfer method was employed to harvest structural brain picture elements to aid CNN. Finally, both types of characteristics were given to the classifier and achieved 79.9% accuracy and 86.1% AUC. This system has low accuracy performance than our work.

Poloni et al. [50] presented a strategy for classifying AD based on hippocampal asymmetry. They analysed the qualities using the asymmetry index (AI), and those that qualified for the analysis of variance test were selectedfor classification. The statistical analysis of AI revealed a substantial enhancement in hippocampal asymmetry as the disease progressed. They achieved 69.44% and 82.59% accuracy and 76% and 90% AUC for MCI vs CN and AD vs CN, respectively. They only performed experiments on combinations of two classes and still achieved lower accuracy than our work.

Goenka et al. [15] suggested the CNN to classify AD and achieved the highest accuracy of 98.3%. Three neuroimaging computational approaches were presented: 3D subject-level, slice-based, and 3D patch-based. The 3D subject-level system was the most accurate among these three approaches and achieved accuracies of 98.26%, 97.48%, and 95.40%, respectively. Even though they have applied three strategies, the performance is less than our study.

Kong et al. [30] created a DL-based approach for AD multi-classification techniques that included a novel MRI, PET image fusion, and 3D CNN to detect AD from an ADNI dataset of 740 3D images and achieved 93.5% accuracy. The performance is less than our study. Poloni et al. proposed two CNNs and achieved a mean absolute error (MAE) of 3.31 and a root means square error (RMSE) of 4.65 for the first approach [49].

In proposed study, the unbalanced MRI dataset was used to detect AD. The dataset consists of 896 MID, 64 MOD, 3200 ND, and 2240 VMD MRI images. Three DCNN models were applied to detect AD but achieved less than 70% accuracy. Furthermore, the four convolutional layer CNN model was implemented and achieved less than 70% accuracy, similar to DCNN models. A new method was tried by combining CNN architectures with the KNN classifier to improve the accuracy and achieved 100% accuracy and 99.58% average accuracy (K = 20). The robustness of the CNN-KNN models was evaluated using the stratified K-fold technique. Various performance metrics, including accuracy, ROC curve, precision, recall, F1-score, MCC, and CKC, were employed to evaluate CNN-KNN. These metric values are above 99%, which shows the model's excellent performance. Even though the dataset is unbalanced, we observed a high rate of AD detection. In the current study, we found a 99.58% average accuracy for K = 20. In comparison with the studies in the literature, our results are remarkable.

The authors examine different CNN network hyperparameter configurations to obtain the best computation complexity and precision to achieve the above results. Variants in input picture size, filter size, optimizers, BN layers, and so on are studied by the author with their own group of hyperparameters. Because of the quantity of filter selection, cost-effective systems may be able to significantly reduce the computer processing cost. The input image size of 224 × 224 has been chosen, and the kernel size of 3 × 3 is selected for the best feature extraction with the Max Pooling layer. This research concentrates on the learning rate context and vector reduction of input data vectors. Various optimizers and batch sizes were also tested, and finally, the Adam optimizer was used with batch size 32. The Optimizer learning rate also varies from 0.1 to 0.00001, and the acceptable accuracy was found at 0.0003. The CNNs flatten layer contains 12,544 selected features with the KNN classifier for categorization.

6 Conclusion

The ability to detect and identify early-stage AD in the healthcare industry is always tricky. The primary objective of this attempt is to promote health competency. In this study, the authors presented a CNN-KNN-based classification method to aid in the detection of Alzheimer's disease. For the first time, the authors demonstrated that CNN-KNN produces many superior outcomes compared to the literature. Although CNN and DCNN have been widely used to analyse brain activity. We merged CNN with KNN and found that this approach has remarkable classification accuracy. CNN-KNN and DCNN were utilised in this study to assess the AD MRI scans of MID, MOD, ND, and VMD patients. In terms of results, the CNN-KNN and DCNN parameters and hyperparameters must be carefully chosen so that the model performs well. First, a CNN model with 16 layers was created, and then three DCNN models (ResNet50, VGG16, and MobileNetV2) were implemented. When the efficiency of these models was compared, it was discovered that the CNN-KNN model performs exceptionally well. The CNN-KNN had an accuracy of 99.58%, a precision of 99.63%, recall of 99.31%, F1-score of 99.43%, MCC of 99.31%, and CKC of 99.31%. On the other hand, the ResNet50, VGG16, and MobileNetV2 models all obtained less than 69% accuracy, 80% precision, 54% recall, and 59% F1-score. These results indicate the CNN-KNN perform very well compared to the DCNN models. The CNN-KNN-based technology will improve physicians' workload and increase the speed of testing methods for Alzheimer's patients. This paper effectively used a CNN-KNN-based structure to classify AD. The CNN KNN performance

was assessed using stratified K-fold, ROC curve, CKC, and MCC, and it was discovered that the model performed admirably.

The results of the present study demonstrate that an MRI image may aid in diagnosing and predicting AD. The experimental results prove that the proposed CNN-KNN architecture is suitable for AD disease detection. The model also achieves outstanding accuracy for four stages of AD classifications. In situations where there are not enough clinical specialists, the proposed method can support an objective supplementary measure for diagnosing AD.

The authors plan to expand on this research to categorise different MRI images from the ADNI and the Open Access Series of Imaging Studies (OASIS) datasets. Based on the information provided, supplemental articles can be written to fill different research gaps. The limitation of this work is that it predicts accurately, but the dataset only contains 64 MRI images for MOD and 896 for MID. The authors plan to use larger datasets with more MRI images in future work. The proposed CNN-KNN model can be deployed to embedded devices and the cloud in the future. The benefit of deploying models to the cloud is that they can be done remotely for AD forecasting. The proposed approach can be improved in the following ways:

- The CNN and ML classifiers, such as SVM, RF, etc., can be integrated so that CNN extracted features can be given to the classifiers.
- The DCNN, like the DenseNet201 model extracted features, can be given to the ML classifiers.
- The Feature selection method can be employed to select CNN extracted features to reduce computing complexity and improve performance.
- The presented model can be developed even more to provide stage-specific AD diagnosis.

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Data availability https://www.kaggle.com/datasets/tourist55/alzheimers-dataset-4-class-of-images

Declarations

Competing interests The authors have no relevant financial or non-financial interests to disclose.

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